Deep Learning Group Project - Bachelor's Degree in Data Science 2026/27

In this project, you will develop a Deep Learning model to perform multiclass image classification.

The dataset used in this project is based on Tiny ImageNet, a reduced image-classification benchmark derived from ImageNet. Tiny ImageNet contains 200 object classes represented by RGB images with a resolution of 64×64 pixels. For this project, a specific version of the dataset was prepared, containing approximately 60,000 images, along with structured metadata, including the filename, class identifier, class description, and numerical label.

The project dataset also includes characteristics commonly found in real-world Machine Learning applications, such as class imbalance and possible label inconsistencies. Students are expected to identify, analyse and, when appropriate, address these characteristics as part of their modelling process.

Your objective is to build a model that predicts the class of a given image based on its visual information.

Like almost every Machine Learning task, there is no single "right solution" for this problem. You can design your model as you wish, as long as you can motivate your choices.

You may apply different preprocessing techniques, image transformations, regularisation methods, hyperparameters and model architectures. At least one approach should use a pre-trained model adapted to the classification task through transfer learning and/or fine-tuning.

The project must be developed in Python 3. You may use libraries such as Pandas, NumPy, Scikit-learn, TensorFlow and Keras. Any external code, pretrained model, library or other resource used must be properly acknowledged.

In addition to your source code (a Python notebook), you should deliver a short scientific report (maximum 5 pages + bibliography, if applicable) containing:

- Group identification, including each member's student number and name.
- A brief description of the dataset and its main characteristics.
- The preprocessing applied to the images and/or labels.
- The model or approaches implemented.
- The experimental setup, including the main Python libraries and implementation details required to reproduce the experiments.
- A brief description of the intermediate models tested, what was changed between experiments and why.
- An explanation of why the final approach was selected.
- An evaluation of your best approach using appropriate metrics.
- A comparison between the main approaches tested.
- A brief error analysis identifying the most common mistakes made by your model.

The report should explain not only what you implemented, but also why the main modelling decisions were taken.

The report must be written in English.

Deliverables

By 11 December 2026, you must deliver:

- the complete source code (***a Python notebook***) used in the project;
- the scientific report in PDF format.

The work must be submitted via Moodle as a single .zip or .rar file named with the group number (for example, GROUP_1.zip). Each group must consist of 5 students. Submissions received up to 24 hours after the deadline will receive a penalty of 3 points. Submissions delivered 24 hours after the deadline will be ignored.

Evaluation

The grade of the project will be obtained by evaluating the following criteria:

- Quality of image/data preprocessing and treatment of data-quality issues - 3 points
- Ability to design and implement Deep Learning models - 3 points
- Ability to use and correctly adapt pre-trained models - 2 points
- Amount and quality of the experimental approaches implemented - 2 points
- Ability to analyse and interpret the results using Deep Learning concepts - 2 points
- Quality of the report and language adequacy - 4 points
- Performance of the model and adequacy of the metrics used - 2 points
- Clarity, organisation and reproducibility of the code - 2 points

Total: 20 points

IMPORTANT: The project will be evaluated on a relative basis. Each group's solution will be assessed not only against the criteria above, but also in comparison with the solutions submitted by the other groups.

Model performance, experimental quality and the quality of the proposed solution will therefore also be considered relative to the results achieved by the remaining groups.

The fact that you are working in a group does not imply that all members of the group will necessarily receive the same grade.

Use of Generative AI Tools

The use of Generative AI tools is permitted as a supporting resource. Any use must be declared in the report, indicating the tool used and the purpose for which it was employed.

The group remains fully responsible for all submitted content, including source code, methodological decisions, results and interpretations. All group members must be able to understand, explain and defend the submitted work, regardless of whether Generative AI tools were used.

Undeclared use of AI-generated content may be treated as a breach of the academic integrity requirements of the project.